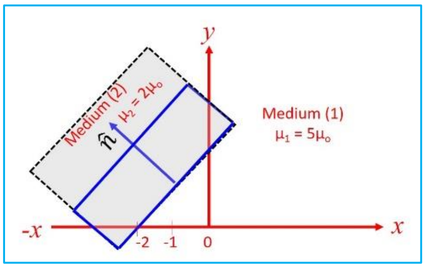
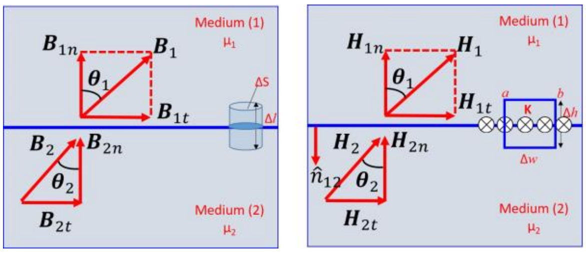
PH 102 (Jan-May-2024), Tutorial-10, Date: 2nd April, 2024 (Tuesday 7:55-8:50 AM)

1. A sphere of radius R has the magnetization $\vec{M} = a\hat{k}\left(\hat{k}.\vec{r}\right)^2$. Determine the following:
   (a) The bound magnetic charge density in the sphere.
   (b) The bound magnetic surface charge density on the sphere.
   (c) The bound surface current density on the sphere.
   (d) The magnetic moment of the sphere.



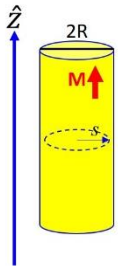
2. (I).The magnetic field intensity in a region-I, $y - x - 2 \le 0$ is $\mathbf{H}_1 = -2\hat{\boldsymbol{x}} + 6\hat{\boldsymbol{y}} + 4\hat{\boldsymbol{z}}$ A/m, where the permeability of region-I is $\mu_1 = 5\,\mu_0$ (see the adjacent figure), calculate the following quantities: (a) Magnetization vector $\mathbf{M}_1$, and Magnetic Flux Density $\mathbf{B}_1$. (b) Magnetic field intensity $\mathbf{H}_2$ and flux density $\mathbf{B}_2$ in the region-II ($y - x - 2 \ge 0$), where the permeability of region-II is $\mu_1 = 2\,\mu_0$.



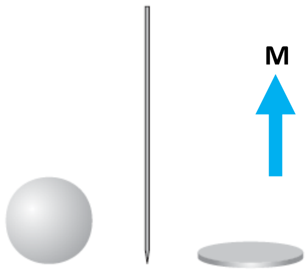
(II). For the boundary between two magnetic media such as is shown in below figure, show that the boundary conditions on the magnetization vector are:

$$\frac{M_{1t}}{\chi_{m1}} - \frac{M_{2t}}{\chi_{m2}} = K \quad \& \quad \frac{\mu_1}{\chi_{m1}} M_{1n} - \frac{\mu_2}{\chi_{m2}} M_{2n}\ .$$

Here χ_{m1} and χ_{m2} are the magnetic susceptibilities of medium **I** and **II**, respectively. If the boundary is not current free show that we obtain the following condition: $\frac{\tan\theta_1}{\tan\theta_2} = \frac{\mu_1}{\mu_2}\left[1 + \frac{K\,\mu_2}{\boldsymbol{B}_2 \sin\theta_2}\right]$

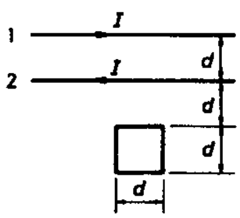


3. An infinitely long cylinder, of radius R, carries a "frozen-in" magnetization, parallel to the axis : $\mathbf{M} = ks\hat{\mathbf{z}}$, where k is a constant and s is the distance from the axis; there is no free current anywhere. Find the magnetic field inside and outside the cylinder by two different methods. (a) First locate all the bound currents, and calculate the field they produce. (b) Use Ampère's law to find H, and then get B (Notice that the second method is much faster, and avoids any explicit reference to the bound currents).



4. Assume the field inside a large piece of magnetic material is $\mathbf{B}_0$, so that $\mathbf{H}_0 = (1/\mu_0)\,\mathbf{B}_0 - \mathbf{M}$, where $\mathbf{M}$ is a "frozen-in" magnetization. Now a small spherical cavity is hollowed out of the material (see adjacent figure). Find the field at the center of the cavity, in terms of $\mathbf{B}_0$ and $\mathbf{M}$. Also find $\mathbf{H}$ at the center of the cavity, in terms of $\mathbf{H}_0$ and $\mathbf{M}$. (b) Find the field long needle-shaped cavity running parallel to $\mathbf{M}$. (c) Find the field for a thin wafer-shaped cavity perpendicular to $\mathbf{M}$.

5. A current **I** flows down a long straight wire of radius a. If the wire is made of linear material (copper, say, or aluminium) with susceptibility χ_m, and the current is distributed uniformly, what is the magnetic field a distance s from the axis? Find all the bound currents. What is the net bound current flowing down the wire?



6. Two infinite parallel wires separated by a distance d carry equal currents I in opposite directions with I increasing at the rate dI/dt. A square loop of wire of length d on a side lies in the plane of the wires at a distance d from one of the parallel wires, as shown in figure below.
   (a) Find the emf induced in the square loop.
   (b) Is the induced current clockwise or anticlockwise? Justify your answer.